

Galileo's telescope

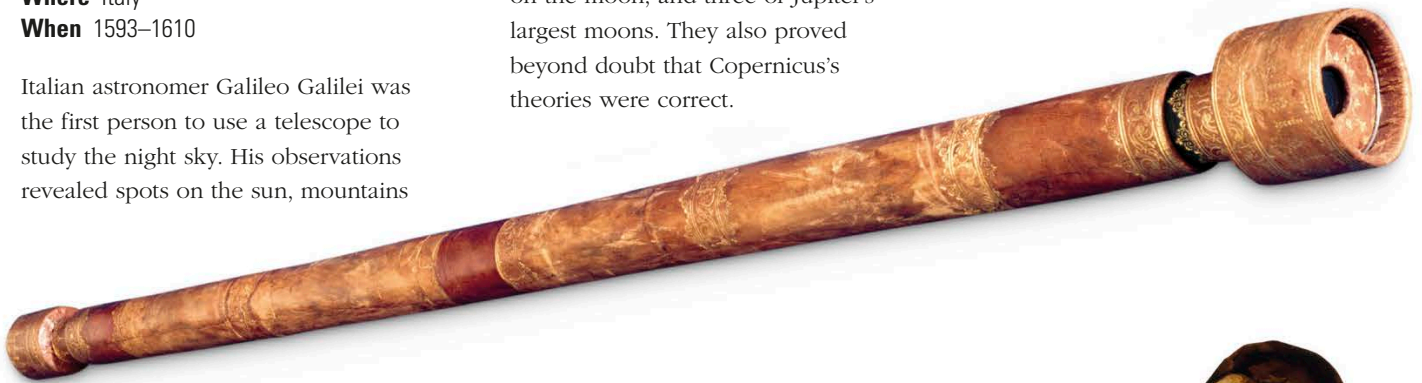
Galileo Galilei

Where Italy

When 1593–1610

Italian astronomer Galileo Galilei was the first person to use a telescope to study the night sky. His observations revealed spots on the sun, mountains

on the moon, and three of Jupiter's largest moons. They also proved beyond doubt that Copernicus's theories were correct.



Scientific method

Francis Bacon

Where England

When 1620

An influential politician, English philosopher Francis Bacon was also a keen scientist. In his book *New Method*, he argued that scientific knowledge was key to improving human life, and that the way to establish scientific facts was by careful observation. His thinking was the basis of the modern scientific method, which uses

experiments to make new discoveries.



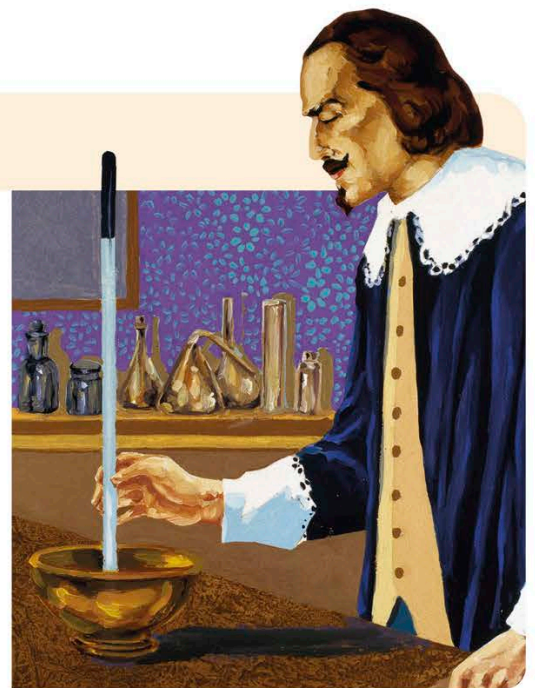
Mercury barometer

Evangelista Torricelli

Where Italy

When 1643

Italian physicist Evangelista Torricelli invented the mercury barometer, the first device that could accurately measure atmospheric pressure—the weight of all the air around us in the atmosphere. Since this pressure changes in different weather conditions, the barometer paved the way for future scientists to develop accurate weather forecasts.



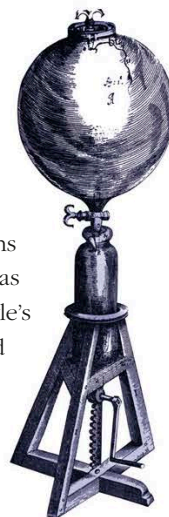
Boyle's law

Robert Boyle

Where England

When 1662

Irish scientist Robert Boyle developed an air pump (right) to study the behavior of gases. This led him to discover Boyle's law, which explains how adding pressure to gas decreases its volume. Boyle's many discoveries, reached through experimentation, helped confirm his status as the first modern chemist.



Laws of motion

Isaac Newton

Where England

When 1687

Celebrated English scientist Isaac Newton was the first to explain how gravity—the invisible force that pulls everything toward the center of the planet—works after watching an apple fall from a tree. He described it in his book *Mathematical Principles of Natural Philosophy*, along with his three laws of motion, which explain the forces that make things move.

